UNIT 5 · TOPIC 2

Physiological Cost of Work and Body Responses

Syllabus topic: Physiological cost and effect of work on physiological responses

Measuring physiological cost

Physiological cost = what the body spends to do the task. It is measured by:

- **OXYGEN CONSUMPTION rate (VO_2, L/min)** — Douglas bag or portable respirometers (Oxylog, K4b² type). Convert to energy: **1 L O_2 ≈ 20.9 kJ ≈ 5 kcal**.
- **HEART RATE (beats/min)** by telemetry — the practical field index.
- Other responses: pulmonary ventilation, blood pressure, body temperature, EMG for muscle load.

Common Indian regression: energy expenditure E (kJ/min) ≈ 0.159 × HR (beats/min) − 8.72; kcal = kJ ÷ 4.187

Work classification and acceptable limits



Indian norms: acceptable workload (AWL) ≈ 35% of VO_2max for an 8-h day; limit of continuous performance ≈ HR 110 beats/min (ΔHR ≤ ~40 over resting)

Christensen grading of physical work — energy expenditure and working heart rate (redrawn from the source figure)

Indian norm (CIAE)	Value
Acceptable workload (AWL), full 8-h day	≈ 35% of the worker's VO_2max
VO_2max	Indian male ~2.0 L/min; female ~1.4–1.7 L/min
Limit of continuous performance	Heart rate ≈ 110 beats/min, or ΔHR (work - rest) ≤ ~40 beats/min

Tasks exceeding these limits need rest pauses or redesign.

Worked example — paddy transplanting.
1. Working HR = 130 beats/min → E = 0.159 × 130 - 8.72 = **11.9 kJ/min = 2.85 kcal/min**.
2. Resting HR = 75 → ΔHR = 130 - 75 = **55 > 40**.
3. The task exceeds the continuous-work limit → provide rest pauses or mechanise (transplanter).
Always state units.

Static vs dynamic work

Work type	Effect
DYNAMIC (rhythmic)	Muscle action pumps blood — sustainable
STATIC (holding)	Sustained contraction occludes blood flow — fatigues quickly even at modest force (holding sprayer lances, clutch pedals, awkward postures)

Design rule: convert static holding into supported or dynamic action — armrests, latching levers, balanced tools.

Practice MCQs

Q13. One litre of oxygen consumed corresponds to about:

(a) 5 kJ (b) 10 kJ
(c) 20.9 kJ (d) 50 kJ

Q14. The practical field index of physiological cost is:

(a) Blood sugar (b) Heart rate (telemetry)
(c) EEG (d) Height

Q15. Oxygen uptake in field studies is measured with:

(a) Anemometer (b) Douglas bag / portable respirometer
(c) Hydrometer (d) Dynamometer

Q16. Using E(kJ/min) = 0.159·HR − 8.72, HR = 130 gives about:

(a) 6 (b) 9
(c) 11.9 (d) 20 kJ/min

Q17. Acceptable workload (AWL) for an 8-h day is about:

(a) 15% (b) 35% of VO_2max
(c) 70% of VO_2max (d) 100%

Q18. The limit of continuous performance corresponds to a working heart rate of about:

(a) 90 (b) 110
(c) 140 (d) 170 beats/min

Q19. The recommended increase of working HR over resting (ΔHR) is at most about:

(a) 10 (b) 25
(c) 40 (d) 80 beats/min

Q20. VO_2max of an average Indian male agricultural worker is about:

(a) 1.0 (b) 2.0
(c) 3.5 (d) 5.0 L/min

Q21. On the Christensen scale, 'moderate' work is:

(a) <2.5 (b) 2.5–5
(c) 5–7.5 (d) 10–12.5 kcal/min

Q22. Work above ~12.5 kcal/min is graded:

(a) Light (b) Moderate
(c) Heavy (d) Extremely heavy

Q23. STATIC muscular work fatigues faster than dynamic because:

(a) It is boring
(b) Sustained contraction occludes blood flow to the muscle
(c) More sweat
(d) Less skill

Q24. Holding a sprayer lance for long periods is an example of:

(a) Dynamic work
(b) Static work
(c) Aerobic exercise
(d) Rest

Q35. Physiological responses to work include:

(a) Heart rate and oxygen uptake
(b) Pulmonary ventilation and body temperature
(c) Muscle EMG activity
(d) All of these

Answers

Q13 — (c) 20.9 kJ
Q14 — (b) Heart rate (telemetry)
Q15 — (b) Douglas bag / portable respirometer
Q16 — (c) 11.9 — *0.159 × 130 − 8.72 = 11.95 kJ/min ≈ 2.85 kcal/min.*
Q17 — (b) 35% of VO_2max
Q18 — (b) 110
Q19 — (c) 40
Q20 — (b) 2.0
Q21 — (c) 5–7.5
Q22 — (d) Extremely heavy
Q23 — (b) Sustained contraction occludes blood flow to the muscle
Q24 — (b) Static work
Q35 — (d) All of these

Sources: Liljedahl, Turnquist, Smith & Hoki — Tractors and their Power Units, Ch. 9 "Human Factors in Tractor Design", pp. 203–246; CIAE Bhopal / AICRP-Ergonomics literature (Gite & Yadav) for Indian physiological norms; SAE J833; ASHRAE comfort work; NPTEL ergonomics lectures.